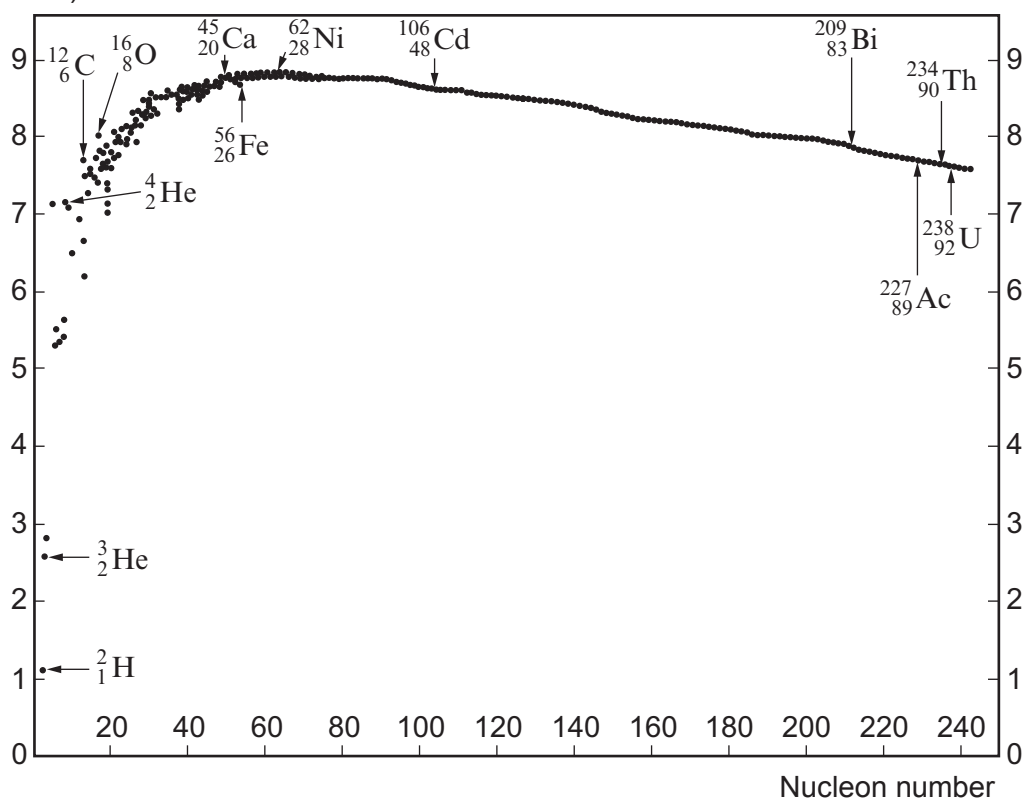
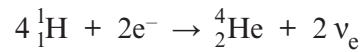


2. (a) Nuclear energy can be released by fusion or fission. Describe these processes by considering the graph of the binding energy per nucleon plotted against the nucleon number. [4]

Binding energy per nucleon
(MeV/nucleon)



- (b) The fusion of hydrogen to helium in the Sun may be represented by:



Particle	Mass / u
${}^1_1\text{H}$	1.00728
${}^4_2\text{He}$	4.00151
e^-	0.00055
ν_{e}	0.00000

$$1\text{ u} = 931\text{ MeV}$$

Calculate the energy released in this reaction in MeV.

[3]

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- (c) Nuclear power has played a role in the generation of electricity in Wales. Wylfa Newydd on Anglesey may feature in further developments. Indicate a benefit and an issue that may arise from such projects and discuss their relative importance. [3]

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